

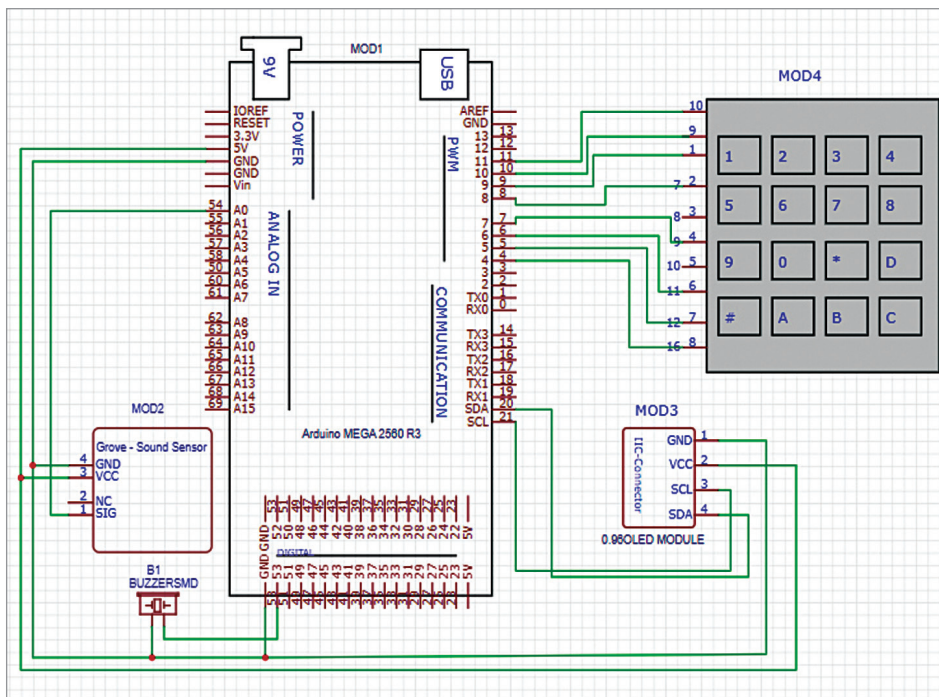


Fig. 2: Circuit diagram

```

sum=0;
for(i=0;i<no_of_samples;i++)
{
  imic_val=analogRead(0);
  sum=sum+imic_val;
}
if ((sum > (sound_high_threshold*no_of_samples))
    && wst_ongoing==false)
{
  no_of_wst_detected++;
  wst_ongoing=true;
}
if ((sum < (sound_low_threshold*no_of_samples))
    && wst_ongoing==true)
{
  wst_ongoing=false;
}
.....
}

```

Fig. 3: A code snippet

Whistle low detection is also important. If the detected flag is true and window sum value is less than the 'low threshold,' whistle low is declared. This section of the C code is shown in Fig. 3.

This method of whistle detection is very robust. It works well even in presence of other household noises.

The other functionalities of software include running the clock timer (minutes and seconds), getting inputs from the 4x4 keypad, displaying data on the OLED, and sounding

the buzzer. The display shows the number of detected whistles and time since start of the cooking process.

Working of the circuit is simple. Before using, set the number of whistles you want and time required for cooking using the keypad. Place the device somewhere near the stove.

When the number of whistles reaches the preset value, the buzzer sounds to indicate that the task of the pressure cooker is over and it is time to turn off the stove. If the pressure cooker does not sound any whistle for a long period of time, then also it sounds the buzzer, after the preset cooking time. This could happen when there is some issue with the pressure cooker because of

which it did not sound the buzzer.

You can upload the code to Arduino using the IDE by selecting the right port of Arduino and board model.

Construction and testing

Upload source code (smart_kitchen_v4.C) of the project on Arduino Mega 2560 board (MOD1). Then wire up the circuit as per the circuit diagram shown in Fig. 2.

Place the device near the cooking stove and, using the keypad, set the number of whistles you want. You may now start the cooking on stove. When the device detects the set number of whistles, it alerts you to turn off the stove.

The author's prototype's external and internal views are shown in Fig. 4. **EFY**

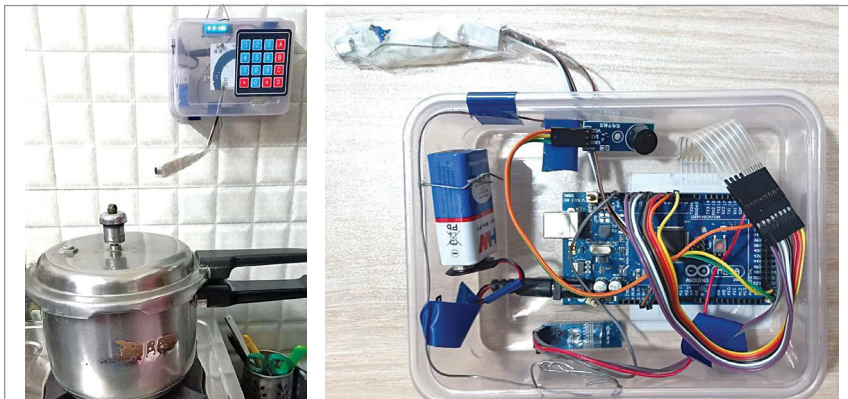


Fig. 4: Author's prototype's external and internal views



Palak Patel is an electronics engineer having experience in embedded system development for terminal automation industry. Currently she is doing freelancing for embedded systems projects